

```

1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s253b IS
6      PORT (
7          clk : IN STD_LOGIC;
8          d_in : IN STD_LOGIC;
9          y : OUT STD_LOGIC_VECTOR(1 DOWNTO 0)
10     );
11 END ENTITY;
12
13 ARCHITECTURE behavioral OF s253b IS
14
15     SIGNAL count : unsigned(1 DOWNTO 0) := "00";
16     SIGNAL readbuf : STD_LOGIC_VECTOR(2 DOWNTO 0) := "000";
17
18 BEGIN
19
20     seq_det_p : PROCESS (clk)
21     BEGIN
22         IF rising_edge(clk) THEN
23
24             -- Check the sequence including the current input bit
25             IF readbuf = "010" THEN
26                 count <= count + 1;
27             END IF;
28
29             -- Shift the current input into the buffer
30             readbuf <= readbuf(1 DOWNTO 0) & d_in;
31
32         END IF;
33     END PROCESS seq_det_p;
34
35     y <= STD_LOGIC_VECTOR(count);
36
37 END ARCHITECTURE behavioral;

```